Yian Yu

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Experience

University of Michigan Michigan, USA
Postdoctoral Research Fellow, Department of Statistics Aug. 2025 – Present
Supervisor: Prof. Yang Chen, Department of Statistics, University of Michigan, Ann Arbor

Education

Southern University of Science and Technology (SUSTech) Shenzhen, China
Ph.D. in Mathematics, Department of Statistics and Data Science Sept. 2020 – June 2025
Supervisor: Prof. Jian Qing Shi, Department of Statistics and Data Science, SUSTech, Shenzhen, China & National Center for Applied Mathematics, Shenzhen, China

SUSTech Shenzhen, China
B.S. in Statistics, Department of Statistics and Data Science Sept. 2016 – June 2020
GPA: 3.74/4.00

University of Oxford Oxford, UK
Summary program at St. Antony's College July-Aug. 2019

Scholarships and Awards

- National Scholarship for Graduate Students, China, 2022. (2%)
- First Prize (Highest Award), 7th National Academic Forum for Doctoral Students in Statistics, China, 2023.
- Top 10 Outstanding Graduate Student, College of Science, SUSTech, 2025. (3%)
- Outstanding Graduating Graduate Student Award, SUSTech, 2025.
- Outstanding Graduate Student Award, SUSTech, 2024.
- Top 10 Outstanding Undergraduate Graduate Award, Department of Statistics and Data Science, SUSTech, 2020.
- Outstanding Undergraduate Thesis Award, SUSTech, 2020. (2%)
- Second Prize, ASC Student Supercomputer Challenge, worldwide, 2018. (Rank 28/300)
- Honorable Mention, Mathematical Contest in Modeling, worldwide, 2018. (24%)
- Second Prize, 14th May Day Mathematical Contest in Modeling, China, 2017.
- First-Class Scholarship for Outstanding Undergraduate Students, SUSTech, 2017. (5%)
- Second-Class Scholarship for Outstanding Undergraduate Students, SUSTech, 2018. (10%)
- National Encouragement Scholarship for Outstanding Undergraduate Students, China, 2019.
- Outstanding Graduate Award, Zhixin College, SUSTech, 2020 (10%)
- Outstanding Student Leader Award, Zhixin College, SUSTech, 2017 and 2018.

Research

Space Weather Forecasting Michigan, USA
Supervisor: Prof. Yang Chen Aug. 2025 – now

- Operational implementation of real-time solar energetic particle event forecasting.
- Applied efficient statistical and machine learning methods for multimodal large-scale spatio-temporal solar data.

Functional Data Analysis Shenzhen, China
Supervisor: Prof. Jian Qing Shi Sept. 2020 – July 2025

- Bayesian regression analysis for dependent data with shape constraints.
- Leveraged a continuous time hidden Markov model for functional data with altered covariance structures.
- Applied on free-living signal data acquired through wearable accelerometers.

Multiple Testing Shenzhen, China
Supervisor: Prof. Siu Hung Cheung, Department of Statistics, The Chinese University of Hong Kong, Hong Kong, China Jan. 2020 – July 2022

- Proposed a modified latent Weibull model, and compared treatments with ordinal responses in trials with sequential monitoring and response-adaptive randomization.

Structural Data Analysis Shenzhen, China
Supervisor: Prof. Jian Qing Shi. And Prof. Yi Guo, Department of Neurology, Shenzhen People's Hospital & Institute of Neurological Diseases, Shenzhen Bay Laboratory, Shenzhen, Guangdong, China Mar. 2022 – Aug. 2023

- Processed questionnaire data, implement statistical tests, confirmatory factor analysis, and logistic regression analysis. Demonstrated the new questionnaire possesses good reliability and validity, outperforming the commonly used scales in terms of screening accuracy.

Programming Skills

Proficient in R. Familiar with Python, MATLAB, and C++.

Publications

(† denotes equal contribution, * denotes corresponding author)

- **Yian Yu**, Yang Chen*, Lulu Zhao, Kathryn Whitman, Ward Manchester, and Tamas Gombosi. (2026). A multi-source dataset for solar energetic particle forecasting. *Scientific Data*, https://doi.org/10.1038/s41597-026-08552-x
- **Yian Yu**, Yang Chen*, Lulu Zhao, Kathryn Whitman, Ward Manchester, and Tamas Gombosi. (2026). Realtime forecasting of solar energetic particle event and proton flux using multi-source solar observations and multi-task deep learning. *Scientific Report*, https://doi.org/10.1038/s41598-026-66110-2
- **Yian Yu**, Yang Chen*, Lulu Zhao, Kathryn Whitman, Ward Manchester, and Tamas Gombosi. (2026). Solar energetic particle forecasting with multi-task deep learning: SEPNET. *Journal of Geophysical Research: Machine Learning and Computation*, 3, e2026JH001247. https://doi.org/10.1029/2026JH001247
- **Yian Yu**, Bo Wang, and Jian Qing Shi*. (2026). Analyzing functional data with a mixture of covariance structures using a curve-based sampling scheme. *Statistics and Computing*, 36 (5), 206 https://doi.org/10.1007/s11222-026-10954-1.
- **Yian Yu**, Long Tang, Kang Ren, Zhonglue Chen, Shengdi Chen, and Jian Qing Shi*. (2024). Bayesian regression analysis for dependent data with an elliptical shape. *Entropy*, 26(12), 1072. https://doi.org/10.3390/e26121072 (The manuscript was awarded the highest prize at the 7th National Academic Forum for Doctoral Students in Statistics)
- Silin Zeng†, **Yian Yu**†, Jian Qing Shi*, Qingshan Geng*, Rafael H Llinas*, Yi Guo*, et al. (2023). Neuro-11: A new questionnaire for the assessment of somatic symptom disorder in general hospitals. *General Psychiatry*, 36, e101082. https://doi.org/10.1136/gpsych-2023-101082 (BMJ Connections journal).
- **Yian Yu**, Cong Xu*, Junjiang Zhong, and Siu Hung Cheung. (2022). Comparison of treatments with ordinal responses in trials with sequential monitoring and response-adaptive randomization. *Statistics in Medicine*, 41(25), 5061-5083. https://doi.org/10.1002/sim.9554

Technical Reports

- **Yian Yu**, Yang Chen*, Lulu Zhao, Kathryn Whitman, Ward Manchester, and Tamas Gombosi. (2026). Realtime forecasting of solar energetic particle event and proton flux using multi-source solar observations and multi-task deep learning. arXiv preprint arXiv:2606.14440. (Submitted, under review)
- **Yian Yu**, Shasha Zou, Tuija Pulkkinen, and Yang Chen*. (2026). Stable Multivariate Functional Time Series Prediction for Major Geomagnetic Indices. arXiv preprint arXiv:2606.14417v1. (Submitted, under review)
- **Yian Yu**, Bo Wang*, and Jian Qing Shi* (2025). Bayesian Latent Factor Models for Multivariate Functional Data with Varying Covariance Structures. (Submitted, under review)
- **Yian Yu**†, Lin Tang†, Jian Qing Shi, Wenjing Zhao*, et al. (2025). Dementia and Parkinson's disease detection using free-living gait data. (Submitted, under review)

Patents

(† denotes equal contribution, * denotes corresponding author)

- Jian Qing Shi, Yutian Zeng, **Yian Yu**, Guanpeng Li, Yingshan Qiu, Jinyu Wang, Huijie Jiang, and Ziyang Ren. (2022). Gait analysis and neurodegenerative disease classification system software. China, 2022SR1591079[CP]. (Application Serial No. 2022R11L2175851, Software Copyright Registration Certificate No. 10545278)
- Jian Qing Shi, Siqi Yan, **Yian Yu**, Beini Liu, Xiaorui Wang, Lin Tang, and Wenjing Zhao. (2025). Neurodegenerative Disease Analysis Method, Device, and Terminal Based on Multimodal Data. China. (Invention Patent Application No. 2025106436652, Current Status: Accepted, pending authorization)

Talks & Posters

- "Prediction Models for Geomagnetic Activity from Functional Data Perspective". Talk at Joint Statistical Meeting (JSM), August 2026, Boston, MA.
- "Prediction Models for Geomagnetic Activity Using Tomorrow's Solar Wind". Poster at Advancing Earth and Space Science (AGU), December 2025, New Orleans, LA.
- "Analyzing Functional Data with a Mixture of Covariance Structure Using a Curve-Based Sampling Scheme". Talk at the 2025 Academic Annual Conference of the Multivariate Analysis Application Committee of China Field Statistics Research Association, June 2025, Chongqing, China.
- "Bayesian Regression Analysis for Elliptical Dependent Variables". Talk at the 2nd Joint Conference on Statistics and Data Science, July 2024, Kunming, China.
- "Bayesian Curve Fitting and Prediction with Ellipsoid-Gaussian Distribution". Talk at the 7th National Academic Forum, Oct. 2023, Guangzhou, China.

Teaching Assistantships

Functional Data Analysis (University faculties), Yunnan University China mathematics Yuan fund summer school for statistics teachers in western universities.	Summer 2023
Probability and Statistics (Undergraduate), SUSTech	Spring 2022
Time Series Analysis (Undergraduate), SUSTech	Fall 2022
Sample Surveys (Undergraduate), SUSTech	Spring 2021
Statistical Linear Models (Undergraduate), SUSTech	Fall 2021